THE SCHEME THEORETIC INTERSECTION OF TWO CLOSED SUBSPACES IS THEIR FIBRE PRODUCT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement displays the cartesian square; the proof handles the closed immersion assertions first and the cartesian property through etale localization.

Editorial note. The two references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Closed subspaces described by sheaves of ideals. Lemma lemma-closed-immersion-ideals (used also at 3213-3215), used at proof lines 2921-2922. Gives the two closed immersion projections.
- Commutation of the scheme theoretic intersection with etale localization. used at proof lines 2923-2925. Reduces the cartesian statement to schemes.
- The scheme case for scheme theoretic intersections. Morphisms, Lemma morphisms-lemma-scheme-theoretic-intersection, used at proof lines 2925-2926. Is the input that closes the argument.

Source reference. Original lemma 03MB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4743–4743. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 0C4I: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15228–15228. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let X be an algebraic space over S. Let $Z, Y \subset X$ be closed subspaces. Let $Z \cap Y$ be the scheme theoretic intersection of Z and Y. Then $Z \cap Y \to Z$ and $Z \cap Y \to Y$ are closed immersions and*

$$\begin{array}{ccc} Z \cap Y & \longrightarrow & Z \\ \downarrow & & \downarrow \\ Y & \longrightarrow & X \end{array}$$

is a cartesian diagram of algebraic spaces over S, i.e., $Z \cap Y = Z \times_X Y$.

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. The morphisms $Z \cap Y \to Z$ and $Z \cap Y \to Y$ are closed immersions by Lemma 03MB. Since formation of the scheme theoretic intersection commutes with étale localization we conclude the diagram is cartesian by the case of schemes. See Morphisms, Lemma 0C4I. □